

AutoEncoder

- neural network that learns a compressed(compact) representation of data
 - architecture
 - encoder: mapping input to latent space
 - decoder: reconstruct the input from the latent code
 - objective
 - minimize reconstruction error
 - deterministic latent representation
 - no explicit probabilistic modeling → not a generative model
 - not define a valid generative distribution
 - random sampling in latent space is unreliable
 - sampling random z does not guarantee meaningful outputs
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Variational AutoEncoder (VAE)

- impose a probabilistic structure on the latent space → learn latent distribution directly
 - encoder
 - $q_\phi(z | x) = \mathcal{N}(\mu(x), \sigma^2(x))$
 - approximate posterior: distribution of latent z given by x
 - output statistics of distribution
 - decoder
 - $p_\theta(x | z)$
 - likelihood: generative distribution of x given by latent z
- **Reparametrization Trick**
 - $z = \mu + \sigma \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$
 - separates randomness from parameters → enabling backpropagation

- objective

- maximizing marginal likelihood of data $\log p(x) \rightarrow$ intractable

latent variable model에서 MLE에 의해 $\log p(x)$ 최대화 $\rightarrow p(x) = \int p(x, z) dz$ posterior $p(z|x) \rightarrow$ intractable
 Variational inference: 잘 알려진 쉬운 분포 $q_\phi(z|x)$ 로 $p(z|x)$ 에 근사 $\log p(x) = \text{ELBO} + D_{KL}(q_\phi(z|x) \| p(z|x))$

ELBO (Evidence Lower Bound) in VAE

ELBO 최대화 $\rightarrow D_{KL}(q_\phi(z|x) \| p(z|x)) \downarrow$ & $\log p(x) \uparrow$

$$\log p(x) \geq \mathbb{E}_{q(z|x)}[\log p(x|z)] - \text{KL}(q(z|x) \| p(z)) \quad \text{ELBO} = \mathbb{E}_{q(z|x)}[\log p(x, z) - q_\phi(z|x)]$$

- instead of maximizing data log likelihood $\rightarrow$ maximizing ELBO $\uparrow = \mathbb{E}_{q(z|x)}[\log p(x|z) + \log p(z) - \log q_\phi(z|x)]$
- **Reconstruction term**
 - encourages accurate reconstruction $\rightarrow$ keep latent informative
- **KL regularization term**
 - pushes posterior toward prior (e.g., Gaussian distribution) $\rightarrow$ smooth $\uparrow$
 - preventing posterior from dirac delta distribution
- drawbacks
 - blurry samples & posterior collapse
 - due to Gaussian likelihood and averaging
 - continuous latent space $\rightarrow$ not ideal for discrete structure

Vector-Quantized Variational AutoEncoder (VQ-VAE)

- replace continuous latent z with discrete codebook entries
 - encoder $\rightarrow$ outputs a vector
 - vector $\rightarrow$ quantized to nearest codebook vector
- objective
 - $\mathcal{L} = \|x - \hat{x}\|^2 + \|\text{sg}[z_e] - e\|^2 + \beta \|z_e - \text{sg}[e]\|^2$
 - reconstruction loss + codebook loss + commitment loss
- captures discrete structure

Generative Model

- push-forward mapping
 - learn a mapping that pushes a simple distribution into a complex data distribution
- two viewpoints:
 - density-based (likelihood): model $p_{\theta}(x)$ directly (e.g., flows)
 - implicit (sampling): model a generator that can sample but may not give tractable likelihood (e.g., GANs)

Evaluation Metrics

- low-level: perceptual metric
- high-level: compare real and generated sample distributions in a feature space
 - metrics often rely on pretrained networks (Inception) as a feature extractor
- caveats
 - not always aligned with human preference
 - sensitive to the feature extractor and dataset domain shift

Peak Signal-to-Noise Ratio (PSNR)

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2$$

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right)$$

- pixel-level reconstruction fidelity
 - based on MSE (mean squared error)
 - MAX: maximum possible pixel value (e.g., 255 or 1)
- higher is better (MSE lower)
- limitation
 - sensitive to small pixel shifts and blur
 - can favor over-smoothed results that look less sharp to humans

Structural Similarity Index (SSIM)

$$\text{SSIM}(x, \hat{x}) = l(x, \hat{x}) \cdot c(x, \hat{x}) \cdot s(x, \hat{x})$$

- similarity of local structure rather than exact pixel match
 - compares luminance, contrast, and structure
 - high-level form
 - typically computed on local windows and averaged
 - often correlates with perceived quality better than PSNR for some tasks
 - higher is better
 - limitation
 - still not a full perceptual metric
 - may not reflect semantic realism in generative outputs
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Fréchet Inception Distance (FID)

$$\text{FID} = |\mu_r - \mu_g|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

- distance between real and generated distributions in Inception feature space
 - approximates each feature distribution as a Gaussian
- extract features (often from Inception pool3)
 - compute mean (quality) and covariance (diversity)
 - real: (μ_r, Σ_r)
 - generated: (μ_g, Σ_g)
- lower is better
- known failure modes
 - biased for small sample sizes (needs enough samples)

- can be gamed if the feature extractor is not appropriate for the domain

Inception Score

$$\text{IS} = \exp \left(\mathbb{E}_x [\text{KL}(p(y|x), p(y))] \right)$$

$$p(y) = \mathbb{E}_x [p(y|x)]$$

- uses a pretrained classifier's outputs $p(y|x)$
 - confident predictions for each image (low entropy per image)
 - quality proxy: $p(y|x)$ is sharp
 - diverse images across the set (high entropy marginal)
 - diversity proxy: $p(y)$ is broad
- higher is better
- limitations
 - does not compare to the real data distribution directly
 - can be high even when samples do not match the target dataset distribution
 - depends strongly on the classifier label space and domain

Learned Perceptual Image Patch Similarity (LPIPS)

$$\text{LPIPS}(x, \hat{x}) = \sum_l w_l |\hat{\phi}_l(x) - \hat{\phi}_l(\hat{x})|_2^2$$

- perceptual distance using deep features from a pretrained network
 - compares images in feature space rather than pixel space
- definition
 - extract multi-layer features $\phi_l(x)$ and $\phi_l(\hat{x})$
 - compute weighted feature differences across layers
 - $\hat{\phi}$: channel-normalized features
- interpretation
 - lower is better (more perceptually similar)

- commonly used for image-to-image translation, restoration, diffusion reconstruction quality
- limitation
 - depends on the backbone and training domain
 - not a distribution metric (unlike FID), it's a pairwise similarity metric

Generative Adversarial Network (GAN)

- implicit generative model (no explicit likelihood)
 - generator $G: z \sim p(z) \rightarrow \hat{x} = G(z)$
 - discriminator D : outputs probability real for an input image
- adversarial training
 - D learns to distinguish real v.s. fake
 - G learns to fool D
 - train as a minimax game
- objective (standard)
 - $\min_G \max_D \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{z \sim p(z)} [\log(1 - D(G(z)))]$
 - common generator variant (stronger gradients)
 - $L_G = -\mathbb{E}_z [\log D(G(z))]$ 학습 초기에 gradient가 약해질 수 있어서
- strengths non-saturating loss ← generator 학습 loss에 관해
 - sharp, high-quality samples discriminator는 제로 (대신 크로스)
 - fast sampling (single forward pass) 잘 학습된 generator를 사용하는게 목표
- weaknesses 실제 data의 분포 $\simeq$ generator가 생성한 sample 분포
 - training instability 결국 해법은 $JS(P_{\text{data}} \parallel P_G) \downarrow$
 - mode collapse (diversity $\downarrow$) ~~~~~
- Wasserstein GAN (WGAN) P_{data} 와 P_G 의 support이 초기에 겹치지 않으면 generator 학습 불안정
 - replace discriminator with critic $f(x)$ (no sigmoid) ↓

optimal transport로 거리 비용도 고려
 (loss가 거리에 가까운 의미가 됨)

- optimize Wasserstein-1 distance proxy $\rightarrow 1 - \text{Lipschitz}$ 가 되도록 critic (discriminator)이
 - $L_D = \mathbb{E}[f(\hat{x})] - \mathbb{E}[f(x)]$ generator가 만든 sample에 관해 critic 함수의 height clipping
 - enforce Lipschitz constraint gradient의 norm이 1이 되도록
 - WGAN-GP: gradient penalty
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Diffusion

- decomposing the mapping into a sequence of small, incremental transformations
 - from a simple distribution (e.g., Gaussian noise)
 - to a complex data distribution
- learn to reverse a gradual noising process
 - instead of directly learning a single complex mapping from noise to data

Denoising Diffusion Probabilistic Models (DDPM)

- probabilistic diffusion model that defines a Markovian forward noising process
 - data distribution $\rightarrow$ gradually corrupted by Gaussian noise $\rightarrow$ normal Gaussian
 - close-form transition in the noising (forward) process
- learns to reverse the noising process step by step using a neural network
 - gradually denoising Gaussian noise estimated by the network (denoiser)

$$q(x_t | x_0) = \mathcal{N}(\sqrt{\alpha_t} x_0, (1 - \alpha_t) \mathbf{I})$$

linear Gaussian Markov chain

Gaussian

$$q(x_{t+1} | x_t, x_0) \approx p_\theta(x_{t+1} | x_t)$$

infinitesimal change $\rightarrow$ Gaussian noise

Gaussian (tractable) $\leftarrow$ approximate

Denoising Diffusion Implicit Models (DDIM)

- assume a non-Markovian forward process
- defines a deterministic mapping from x_t to x_{t-1}
 - prediction $\rightarrow$ decomposed into clean data x_0 + noise vector to $t - 1$ (+ stochastic term)